

KENDRIYA VIDYALAYA SANGATHAN, BHUBANESWAR REGION

SESSION:2024-25

PRE BOARD-I

MAHEMATICS-BASIC(241)

DURATION: 3 Hrs.

F.M: 80Mark

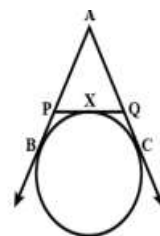
General Instructions:

Read the following instructions carefully and follow them:

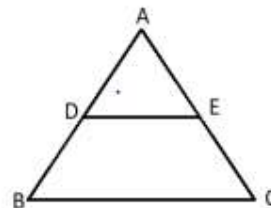
1. This question paper contains 38 questions.
2. This Question Paper is divided into 5 Sections A, B, C, D and E.
3. In Section A, Questions no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.
4. In Section B, Questions no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
5. In Section C, Questions no. 26-31 are short answer (SA) type questions, carrying 03 marks each.
6. In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 05 marks each.
7. In Section E, Questions no. 36-38 are case study based questions carrying 4 marks each with sub parts of the values of 1, 2 and 1 marks each respectively.
8. All Questions are compulsory. However, an internal choice in 2 Questions of section B, 2 Questions of section C and 2 Questions of section D has been provided. And internal choice has been provided in all the 2 marks questions of Section E.

SECTION-A

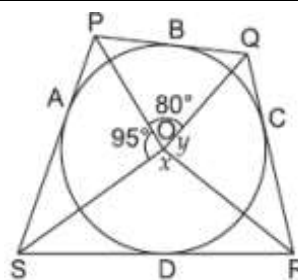
1.	If two positive integers a and b are written as $a = x^4y^2$ and $b = xy^3$, where x and y are prime numbers, then HCF(a, b) is: (A) xy^3 (B) x^4y^3 (C) xy^2 (D) xy^2	1
2.	If in $\triangle ABC$ and $\triangle DEF$, $\frac{AB}{DE} = \frac{BC}{FD}$, then triangles will be similar, if (A) $\angle A = \angle F$ (B) $\angle A = \angle D$ (C) $\angle B = \angle D$ (D) $\angle B = \angle E$	1
3.	In the adjoining figure, if $AB=28$ cm, then find the perimeter of $\triangle APQ$. (A) 50 cm (B) 56 cm (C) 60 cm (D) 64 cm	1
4.	If the volume of hemisphere is $18\pi \text{ cm}^3$, then its diameter is (A) 12 cm (B) 3 cm (C) 4.5 cm (D) 6 cm	1
5.	Which of the following cannot be the probability of an event? (A) 1.5 (B) $\frac{3}{5}$ (C) 25% (D) 0.589	1
6.	The ratio between LCM and HCF of 5, 15, 20 is: (A) 9:1 (B) 4:3 (C) 11:1 (D) 12:1	1

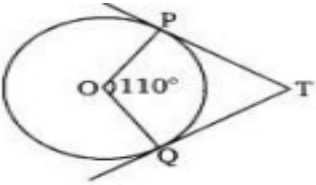


7.	If $\cos A = \frac{1}{2}$, then the value of $(\cos A - \sec A)$ is (A) $\frac{3}{2}$ (B) $-\frac{3}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $-\frac{\sqrt{3}}{2}$	1														
8.	Find the value of K for which the equation $4x^2 - 8Kx + 9 = 0$ has equal roots. (A) $\frac{1}{2}$ (B) 0 (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{12}{8}$	1														
9.	According to Basic Proportionality Theorem which of the following is true? (A) $\frac{AB}{BD} = \frac{AE}{EC}$ (B) $\frac{AD}{BD} = \frac{AE}{EC}$ (C) $\frac{AD}{AB} = \frac{AC}{AE}$ (D) All of these	1														
10.	If $\left(\frac{a}{2}, 4\right)$ is the mid-point of the line segment joining the points A(-6,5) and B(-2,3) then the value of a is (A) - 8 (B) 3 (C) - 4 (D) 4	1														
11.	For the following distribution: the lower limit of the median class is <table border="1"><thead><tr><th>Marks</th><th>Below 10</th><th>Below 20</th><th>Below 30</th><th>Below 40</th><th>Below 50</th><th>Below 60</th></tr></thead><tbody><tr><td>No of students</td><td>3</td><td>12</td><td>27</td><td>57</td><td>75</td><td>80</td></tr></tbody></table> (A) 20 (B) 30 (C) 40 (D) 50	Marks	Below 10	Below 20	Below 30	Below 40	Below 50	Below 60	No of students	3	12	27	57	75	80	1
Marks	Below 10	Below 20	Below 30	Below 40	Below 50	Below 60										
No of students	3	12	27	57	75	80										
12.	Which of the following is not defined? (A) $\sec 0^\circ$ (B) $\operatorname{cosec} 90^\circ$ (C) $\tan 90^\circ$ (D) $\cot 90^\circ$	1														
13.	In the given figure, if quadrilateral PQRS circumscribes a circle, then (A) $x = 95^\circ, y = 95^\circ$ (B) $x = 100^\circ, y = 85^\circ$ (C) $x = 110^\circ, y = 90^\circ$ (D) $x = 85^\circ, y = 90^\circ$	1														
14.	If a pair of linear equations in two variables is consistent, then the lines represented by two equations are (A) intersecting (B) parallel (C) always coincident (D) Both Intersecting and coincident	1														
15.	Find 100 th term of the AP: 4 , 9, 14, (A) 495 (B) 499 (C) 504 (D) 49	1														



Marks	Below 10	Below 20	Below 30	Below 40	Below 50	Below 60
No of students	3	12	27	57	75	80



16.	The roots of the quadratic equation $x^2 - 2x + 1 = 0$ are (A) 1, 1 (B) 2, 2 (C) -1, -1 (D) 1, -1	1
17.	The least number that is divisible by all the numbers 1 to 10 (both inclusive) is (A) 10 (B) 504 (C) 2520 (D) 100	1
18.	In figure, if TP and TQ are the two tangents to a circle with centre O so that $\angle POQ = 110^\circ$, then $\angle PTQ$ is equal to (A) 60° (B) 70° (C) 80° (D) 90° 	1
19.	DIRECTION: In the question numbers 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (b) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A). (c) Assertion (A) is true but Reason (R) is false. (d) Assertion (A) is false but Reason (R) is true. Assertion(A): The zeroes of the polynomial $p(x) = (x-1)(x-2)(x-3)$ are 1, 2 and 3. Reason(R): The zeroes of a polynomial are the x-coordinates of the points where the graph of the polynomial intersects x-axis or the points on the graph where $p(x)=0$.	1
20.	Assertion(A): 50 th term of the sequence 3, 7, 11, is 184 Reason(R): The nth term of an A.P. is given by $a_n = a + (n - 1)d$	1

SECTION-B

21.	Evaluate : $\frac{2 \sin^2 60^\circ - \tan^2 30^\circ}{\sec^2 45^\circ}$	2
22.	(x, y) is a point which is equidistant from the points A(4,3) and B(3,4). Prove that $x - y = 0$. OR Find the coordinates of point P which divides the line segment joining the points A (-2,5) and B (3,-5) in the ratio 2:3.	2
23.	Prove that the lengths of tangents drawn from an external point to a circle are equal.	2
24.	The sum of the first 12 terms of an A.P. is 900. If its first term is 20, then find the common difference and 12th term. OR Determine the A.P. whose 3 rd term is 5 and 7 th term is 9.	2

25.

Find the median of the following frequency distribution table.

Class interval	10 - 25	25 - 40	40 - 55	55 - 70	70 - 85	85 - 100
Number of students	2	3	7	6	6	6

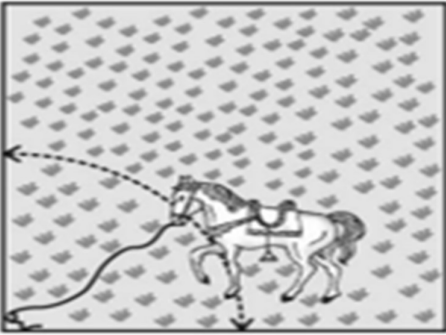
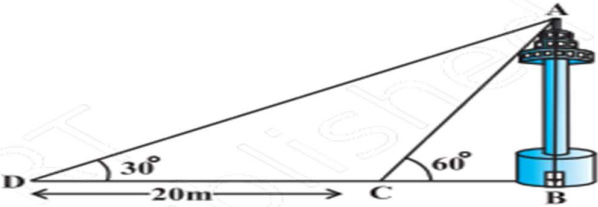
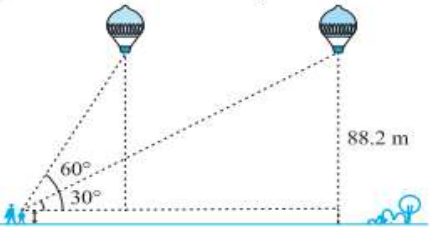
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SECTION-C

26.	Prove that $\sqrt{5}$ is an irrational number.	3														
27.	Prove that: $(\operatorname{Cosec} \theta - \cot \theta)^2 = \frac{1 - \cos \theta}{1 + \cos \theta}$	3														
28.	If (1, 2), (4, y), (x, 6) and (3, 5) are the vertices of a parallelogram taken in order, find x and y. OR Find the coordinates of the points P and Q which trisects the line segment AB joining the points A(-2, 2) and B (7, -4).	3														
29.	Solve the following pair of equations by substitution method and find x and y: $7x - 15y = 2$ $x + 2y = 3$	3														
30.	The distribution below shows the number of wickets taken by bowlers in one-day cricket matches. Find the mean number of wickets by choosing a suitable method. What does the mean signify? <table border="1"><thead><tr><th>Number of wickets</th><th>20 - 60</th><th>60 - 100</th><th>100 - 150</th><th>150 - 250</th><th>250 - 350</th><th>350 - 450</th></tr></thead><tbody><tr><th>Number of bowlers</th><td>7</td><td>5</td><td>16</td><td>12</td><td>2</td><td>3</td></tr></tbody></table>	Number of wickets	20 - 60	60 - 100	100 - 150	150 - 250	250 - 350	350 - 450	Number of bowlers	7	5	16	12	2	3	3
Number of wickets	20 - 60	60 - 100	100 - 150	150 - 250	250 - 350	350 - 450										
Number of bowlers	7	5	16	12	2	3										
31.	A quadrilateral ABCD is drawn to circumscribe a circle . Prove that : $AB + CD = AD + BC$ OR Prove that opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle.	3														

SECTION-D

32.	Prove the Basic proportionality theorem: If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.	5
33.	Find two consecutive positive integers, the sum of whose squares is 365. OR A motor boat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.	5

34.	A horse is tied to a peg at one corner of a square shaped grass field of side 15 m by means of a 5 m long rope (see Fig.). Find (i) the area of that part of the field in which the horse can graze. (ii) the increase in the grazing area if the rope were 10 m long instead of 5 m. (Use $\pi = 3.14$)		5
35.	A TV tower stands vertically on a bank of a canal. From a point on the other bank directly opposite the tower, the angle of elevation of the top of the tower is 60°. From another point 20 m away from this point on the line joining this point to the foot of the tower, the angle of elevation of the top of the tower is 30° (see Fig.). Find the height of the tower and the width of the canal. OR A 1.2 m tall girl spots a balloon moving with the wind in a horizontal line at a height of 88.2 m from the ground. The angle of elevation of the balloon from the eyes of the girl at any instant is 60°. After some time, the angle of elevation reduces to 30° (see Fig.). Find the distance travelled by the balloon during the interval.	 OR 	5

SECTION-E

36.	Ram and Imran went to a fair and they want to try their luck at a shop where two dice are thrown at once to win various exclusive gifts . (i) Find the probability of getting a total of 10 on both the dice. (ii) The shop keeper announces a gift of a refrigerator if any one got different numbers on both the dice. Find the probability that Ram throws the dice and “not getting same number on two dice”. OR Find the probability that Imran will get prime numbers on both the dice. (iii) Find the probability of getting odd numbers on both dice.		1 2 1
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37.	A gulab jamun, contains sugar syrup up to about 30% of its volume. A shopkeeper prepares 45 special gulab jamuns for his family friends who are coming to their house on eve of Diwali each shaped like a cylinder with two hemispherical ends with length 5 cm and diameter 2.8 cm (see Fig.) (i) Find the height of the cylindrical part of the gulab jamun. (ii). Find approximately how much syrup would be found in 45 gulab jamuns. OR Find how much space will be occupied by all the 45 gulab jamuns if they are kept in a cylindrical container as shown in the figure. (iii)What will be the volume of a laddoo (spherical in shape) if its radius is 1cm.	1 2 1
38.	Pankaj's father gave him some money to buy avocado from the market at the rate of $p(x) = x^2 - 24x + 128$. Let α and β are the zeroes of $p(x)$ and $\alpha > \beta$.  Based on the above information, answer the following questions. (i) Find the zeroes of polynomial $p(x) = x^2 - 24x + 128$. (ii)Form a polynomial whose zeroes are $(\alpha+2)$ and $(\beta-1)$ given $\alpha = 3, \beta = -2$ OR If -3 is one of the zero of the polynomials $kx^2 + 6x + 12$, then find the value of k . (iii) Find the value of $(\alpha + \beta)^2$ if $p(x) = x^2 - 24x + 128$.	1 2 1